

XLK Microstructure Statistical Arbitrage: Final-Data Progress Report

Market-neutral XLK-vs-basket arbitrage is audited separately from XLK-only sparse-basket fair-value timing. The report emphasizes spread construction, gross/no-cost economics, transaction-cost scenarios, and last-trade execution proxies.

Top Holdings

symbol	official_weight_pct	basket_weight
NVDA	14.4920	0.1861
AAPL	12.3596	0.1587
MSFT	9.2484	0.1187
AVGO	6.0025	0.0771
MU	4.4668	0.0574
AMD	4.3026	0.0552
INTC	3.4237	0.0440
CSCO	2.6563	0.0341
PLTR	2.4089	0.0309
LRCX	2.3465	0.0301
AMAT	2.2604	0.0290
ORCL	2.1327	0.0274
TXN	1.8667	0.0240
KLAC	1.6563	0.0213
IBM	1.5886	0.0204
ADI	1.4224	0.0183
QCOM	1.3824	0.0177
ANET	1.3053	0.0168
SNDK	1.2825	0.0165
APH	1.2803	0.0164

Data Diagnostics

symbol	minutes	median_spread_bps	trade_count
AAPL	89036	2.55	139411801
ADI	69042	23.04	11433530
AMAT	73824	18.56	21025743

symbol	minutes	median_spread_bps	trade_count
AMD	86676	6.83	115425809
ANET	76955	14.75	22352603
APH	83767	8.25	21496428
AVGO	81264	10.16	81720848
CSCO	94285	2.32	37859860
IBM	74934	11.11	20050703
INTC	95223	4.31	97876707
KLAC	70671	44.18	7614350
LRCX	79947	9.45	29280656
MSFT	84612	5.86	115802312
MU	83149	8.96	79375851
NVDA	95743	1.49	495268838
ORCL	78274	9.39	68226541
PLTR	90375	4.27	155666878
QCOM	78211	10.11	28969759
SNDK	73310	34.21	29732564
TXN	75598	15.07	19836036
XLK	97400	1.07	49165476

Professor Cost/Spread Leaderboard

pair	spread_type	no_cost_midpoint	quarter_spread_cost	half_spread_taker_cost	clipped_last_trade_proxy
XLK-AMD	direct_log_price	453.35	233.95	14.55	383.05
XLK-NVDA	direct_log_price	409.08	147.99	-113.09	280.18
XLK-PLTR	price_regression_residual	254.29	126.01	-2.27	244.60
XLK-AAPL	cum_residual_return	200.10	113.11	26.13	182.43
XLK-INTC	direct_log_price	52.23	25.81	-0.61	67.62
XLK-MSFT	direct_log_price	102.43	-10.87	-124.17	97.12
XLK-ORCL	price_regression_residual	1.61	-61.62	-124.85	-48.59
XLK-ANET	direct_log_price	88.01	-104.22	-296.45	116.26
XLK-CSCO	cum_residual_return	-67.95	-111.05	-154.16	-43.99
XLK-KLAC	price_regression_residual	562.05	-120.52	-803.08	517.02
XLK-ORCL	cum_residual_return	-53.21	-123.52	-193.83	-45.10
XLK-LRCX	price_regression_residual	174.01	-135.42	-444.85	165.69

Monetization Optimizer

decision	candidate_set	validation_net_bps	test_net_bps	test_max_drawdown_bps
no_trade	liquid_validation	526.14	-712.26	-1393.79

Sparse Bid/Ask Audit

sample	old_net_bps	new_bidask_cost_bps	new_bidask_net_bps
test	2.26	29.01	-17.32
train	-178.81	669.92	-593.42

Robust Alpha Selection

decision	economic_label	selected_strategy	train_net_bps	test_net_bps
active_rule	XLK-only timing	xlk_only_timing_micro_conf_0.5_ridge_pos_k3_zscore_mean_e3_x0.5_h0_contra_nogate_cw1950_mh180	335.69	-202.21

Top-20 Method Diagnostics

method	selected_pairs	gate_pass_pairs	raw_test_net_bps	gate_test_net_bps	test_trades
baseline	20	0	-14635.66	0.00	2848
coint_filter	3	0	-837.03	0.00	322
combined	3	0	-979.44	0.00	218
kalman_lagged	20	0	-12823.38	0.00	2609
passive_stress	20	0	-9767.92	0.00	2943
sscore_ou_gate	20	0	-12309.88	0.00	2513

No-Trade Gate Examples

pair_or_basket	method	validation_net_bps	test_net_bps	gate_selected_flag	gate_reason
XLK-AAPL	baseline	144.27	-288.97	False	2x_cost_net<0
XLK-AAPL	kalman_lagged	-23.38	-38.05	False	validation_net<=0
XLK-AAPL	passive_stress	143.86	-233.62	False	2x_cost_net<0
XLK-AAPL	sscore_ou_gate	-391.96	-341.26	False	validation_net<=0
XLK-ADI	baseline	-300.53	-785.40	False	validation_net<=0
XLK-ADI	kalman_lagged	-310.80	-888.98	False	validation_net<=0
XLK-ADI	passive_stress	-208.78	-615.98	False	validation_net<=0
XLK-ADI	sscore_ou_gate	-337.25	-1051.03	False	validation_net<=0

pair_or_basket	method	validation_net_bps	test_net_bps	gate_selected_flag	gate_reason
XLK-AMAT	baseline	-13.39	-2463.93	False	validation_net<=0
XLK-AMAT	kalman_lagged	-146.00	-1588.53	False	validation_net<=0

Empirical Symbol Costs

symbol	median_spread_bps	p90_spread_bps	median_halfspread_bps	median_volume
XLK	1.07	9.51	0.53	16103.00
NVDA	1.49	7.68	0.74	289739.65
CSCO	2.32	6.75	1.16	22915.82
AAPL	2.55	21.82	1.27	62436.51
PLTR	4.27	21.95	2.14	93897.57
INTC	4.31	8.98	2.16	142208.00
MSFT	5.86	34.49	2.93	31149.00
AMD	6.83	33.16	3.42	69868.00
APH	8.25	36.14	4.13	10533.00
MU	8.96	32.87	4.48	39431.00
ORCL	9.39	35.47	4.70	29401.00
LRCX	9.45	37.50	4.72	13180.45
QCOM	10.11	32.21	5.06	10909.00
AVGO	10.16	38.71	5.08	30074.27
IBM	11.11	35.24	5.56	5576.00
ANET	14.75	47.30	7.38	11351.00
TXN	15.07	41.27	7.53	7682.24
AMAT	18.56	50.14	9.28	8285.27
ADI	23.04	52.50	11.52	3930.00
SNDK	34.21	76.25	17.10	11535.00

Latency Taker Costs

symbol	latency_min	buy_taker_cost_bps_median	sell_taker_cost_bps_median
AAPL	0	1.27	1.27
AAPL	1	2.19	2.23
AAPL	2	2.39	2.43
AAPL	5	2.72	2.73
ADI	0	11.52	11.52

symbol	latency_min	buy_taker_cost_bps_median	sell_taker_cost_bps_median
ADI	1	11.32	11.29
ADI	2	11.42	11.39
ADI	5	11.65	11.32
AMAT	0	9.28	9.28
AMAT	1	9.34	9.17
AMAT	2	9.56	9.21
AMAT	5	10.06	9.23
AMD	0	3.42	3.42
AMD	1	4.51	4.40
AMD	2	4.78	4.56
AMD	5	5.23	4.82
ANET	0	7.38	7.38
ANET	1	7.70	7.61
ANET	2	8.09	7.79
ANET	5	8.64	7.90

Execution-Optimized Pair Audit

stock	spread_type	policy	val_net_bps	test_net_bps	oos_decision	final_policy
INTC	direct_log_price	taker	2883.13	-3253.43	reject_after_oos_audit	no_trade

Narrow Rescue Pair Gate

pair	spread_type	exact_bidask_latency0	exact_bidask_latency1	exact_bidask_latency5	clipped_last_trade_proxy	passes_rescue_gate
XLK-AAPL	cum_residual_return	26.12	1.47	-33.37	182.43	True
XLK-AMD	direct_log_price	14.53	-266.99	-308.30	383.04	False
XLK-INTC	direct_log_price	-0.63	17.85	-10.62	67.63	False
XLK-PLTR	price_regression_residual	-2.25	-143.84	-365.55	244.59	False

No-Short Rescue Audit

selection_type	decision	strategy	train_net_bps	validation_net_bps	test_net_bps	test_2x_cost_net_bps
walk_forward_selected	no_trade					
test_oracle_not_selectable	diagnostic_only	short_uptrend_multi_day_trend_lb3900_thr200_flat	-3268.65	24.54	-37.71	-277.45
predefined_reference	diagnostic_only	short_uptrend_multi_day_trend_lb780_thr0_flat	-3797.60	-18.59	-302.40	-642.94

selection_type	decision	strategy	train_net_bps	validation_net_bps	test_net_bps	test_2x_cost_net_bps
predefined_reference	diagnostic_only	short_uptrend_multi_day_trend_lb780_thr50_flat	-3834.25	16.74	-534.29	-1018.50
predefined_reference	diagnostic_only	short_uptrend_multi_day_trend_lb390_thr50_flat	-4401.37	51.27	-940.53	-1463.16
predefined_reference	diagnostic_only	short_uptrend_intraday_trend_lb120_thr25_flat	-5110.47	-174.96	-1403.33	-2261.63
predefined_reference	diagnostic_only	short_uptrend_multi_day_trend_lb390_thr0_flat	-5136.82	-230.82	-956.93	-1598.53
predefined_reference	diagnostic_only	short_uptrend_intraday_trend_lb120_thr0_flat	-6100.82	-22.10	-1409.26	-2362.22

Loss Streamline Decision

research_path	evidence_tier	policy_role	decision	test_net_bps
market_neutral_pair_or_basket	final_12m_evidence	selected_no_trade_policy	no_trade	0.00
robust_alpha_suite_selected	final_12m_evidence	selected_no_trade_policy	no_trade	-202.21
fixed_bps_xlk_only_timing_candidate	legacy_candidate_screen	not_policy_eligible	legacy_candidate_shape_only	613.35
finaldata_fixed_bps_xlk_only_timing	final_12m_evidence	selected_no_trade_policy	no_trade	-977.24
timing_robustness_current_selection	final_12m_evidence	selected_no_trade_policy	no_trade	0.00

Finaldata Fixed-BPS Timing Controls

decision	basket_symbols	train_net_bps	validation_net_bps	test_net_bps	circular_pvalue
no_trade	NVDA AAPL MSFT AVGO MU	-2572.04	-45.13	-977.24	0.66

Microstructure Refinement Horizon Sweep

horizon_min	decision	train_net_bps	validation_net_bps	test_net_bps	test_trades
5	no_trade	0.00	0.00	0.00	0.00
15	no_trade	0.00	-44.37	0.00	0.00
30	no_trade	-39.94	-40.85	0.39	14.00
60	no_trade	-14.90	-213.33	-47.30	40.00

Current Timing Robustness Re-Audit

decision	reason	train_net_bps	test_net_bps
no_trade	No XLK-only timing rule passed train/validation filters.	0.00	0.00

Named Timing Candidate Re-Audit

strategy	basket_symbols	train_net_bps	mar_net_bps	apr_net_bps	test_net_bps
micro_shrink_0.75_cw10d_e60_x0_mh240	NVDA AAPL MSFT AVGO MU	-2695.47	-32.45	-862.35	-894.79

Regime Shift Summary

metric	train_avg	test_avg	test_minus_train
xlk_basket_corr_1m	0.61	0.63	0.02
xlk_on_basket_beta_1m	0.53	0.62	0.09
residual_vol_bps_1m	4.22	5.12	0.91
median_xlk_spread_bps	1.13	1.02	-0.11
signal_std_bps	351.91	150.39	-201.52
alpha_ic_5m	0.00	0.03	0.03
contrarian_decile_edge_5m	0.33	2.38	2.05
alpha_ic_15m	-0.01	0.04	0.05
contrarian_decile_edge_15m	0.20	5.88	5.68
alpha_ic_30m	-0.01	0.05	0.06
contrarian_decile_edge_30m	0.24	10.24	10.00
alpha_ic_60m	-0.01	0.08	0.09

Regime-Gate Repair Selection

decision	selected_strategy	gate_mode	state_kind	train_net_bps	test_net_bps	test_2x_cost_net_bps
diagnostic_only	two_sided_premium_persistence_lb780_thr75_flat	two_sided	premium_persistence	79.63	-131.36	-210.74

Regime Classifier Selection

decision	selected_strategy	train_scheme	model_name	validation_net_bps	test_net_bps	test_2x_cost_net_bps
diagnostic_only	train_all_rf_depth4_h60_edge5_p0.4	train_all	rf_depth4	104.32	-271.67	-931.31

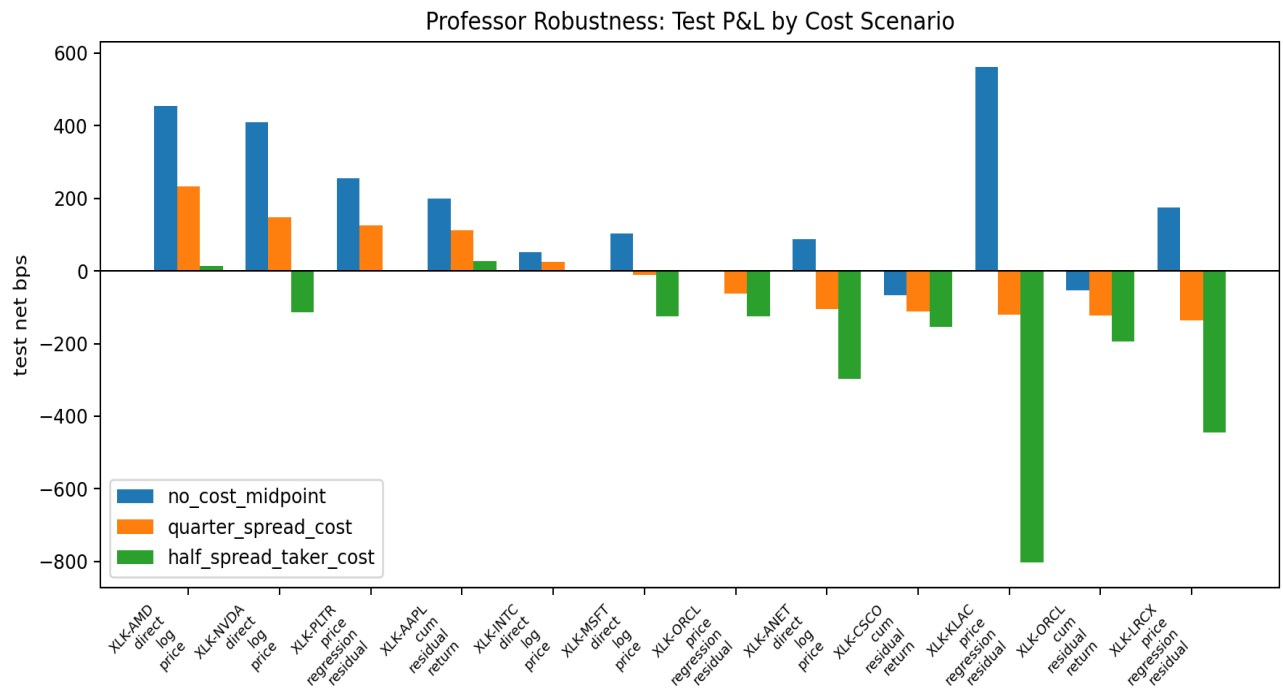
Robust Alpha Controls

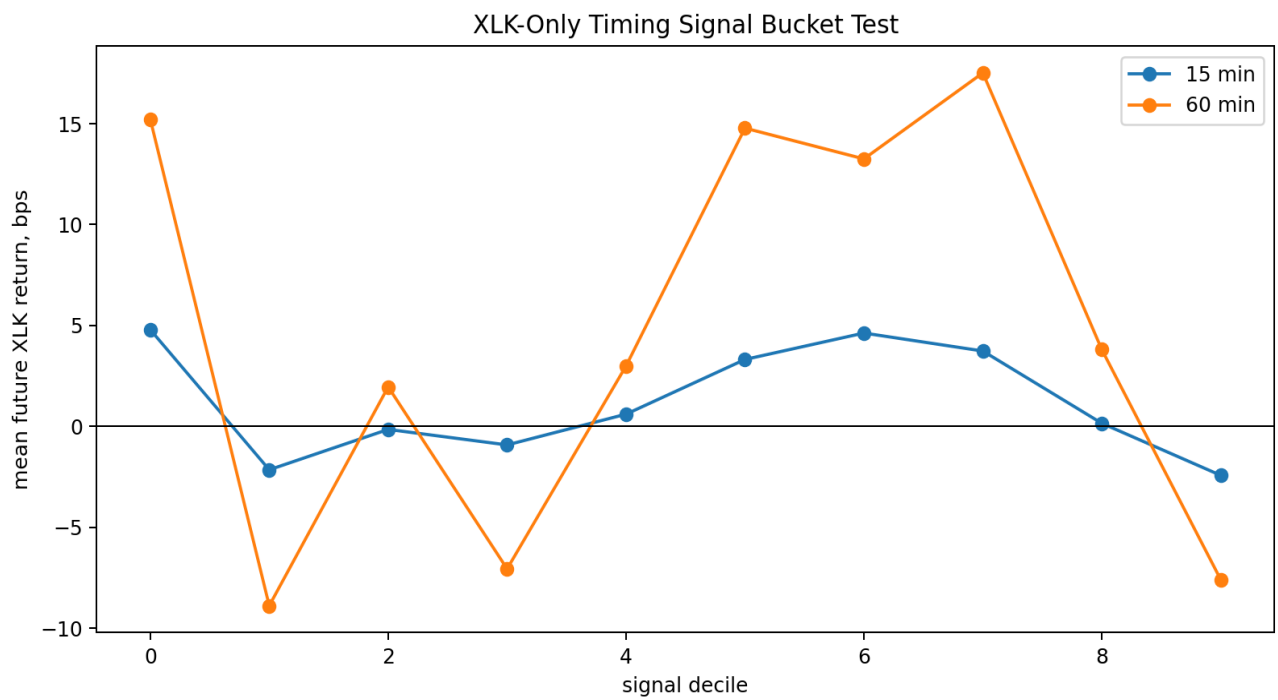
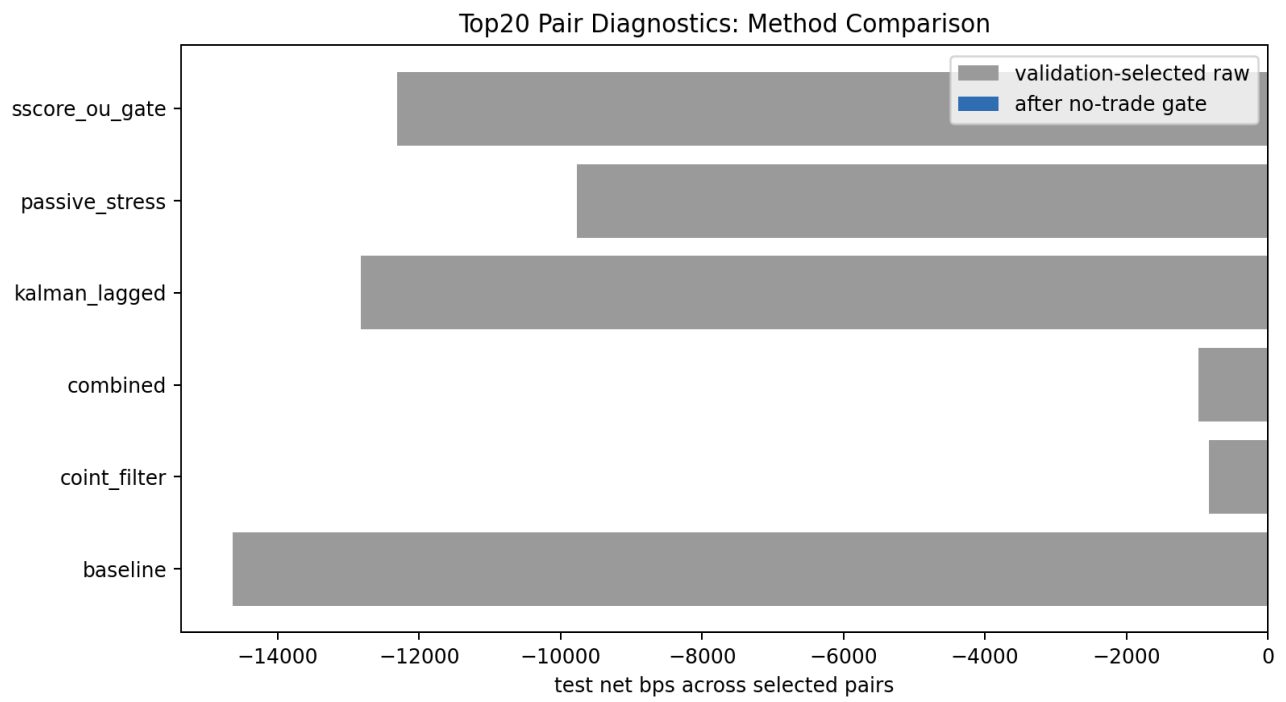
control	train_net_bps	test_net_bps	test_trades
selected	335.69	-202.21	16.00
sign_flip	-595.83	170.69	16.00
active_always_long	160.08	81.08	16.00
active_always_short	-420.22	-112.60	16.00

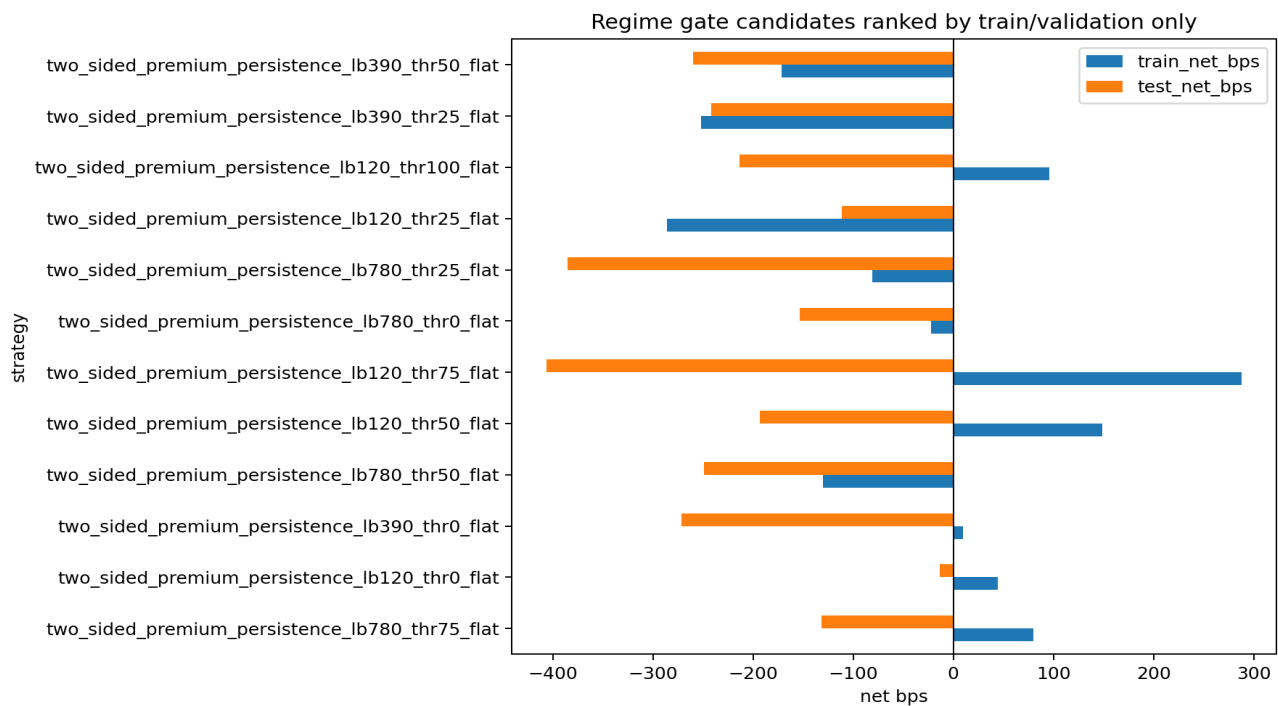
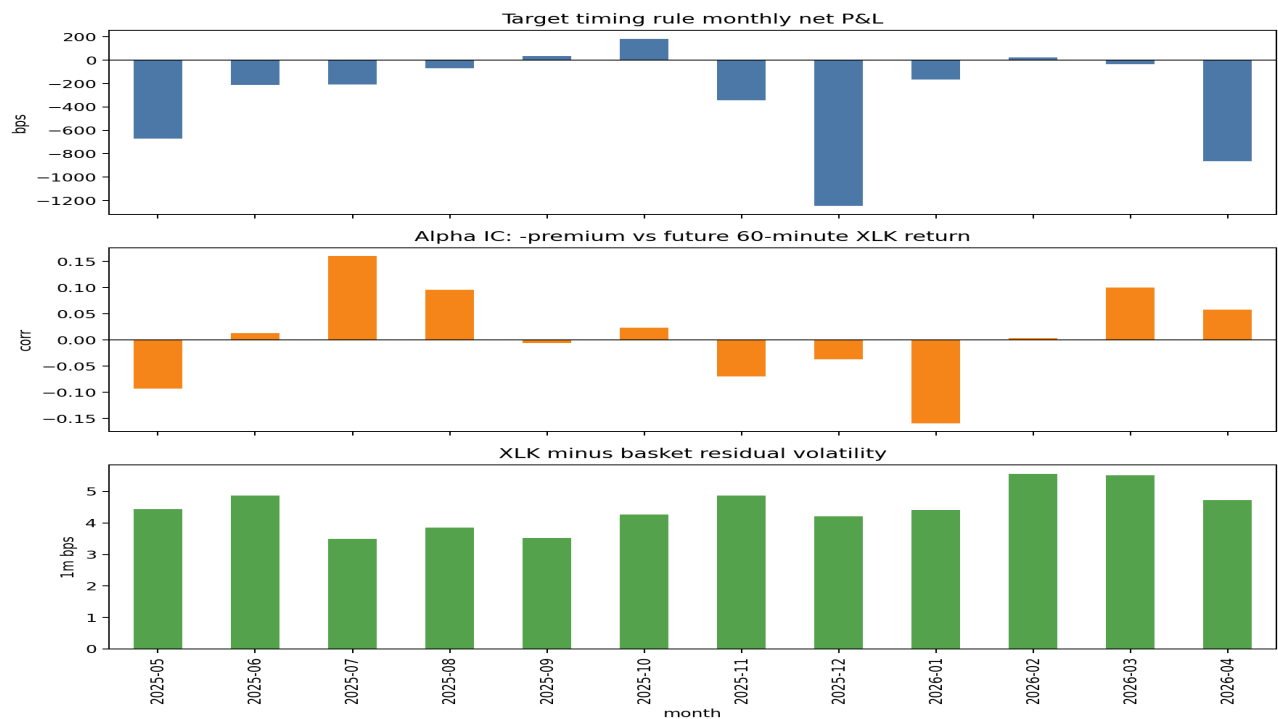
control	train_net_bps	test_net_bps	test_trades
circular_shift_mean		-251.46	
circular_shift_p95		-125.15	
selected_vs_circular_pvalue		0.34	

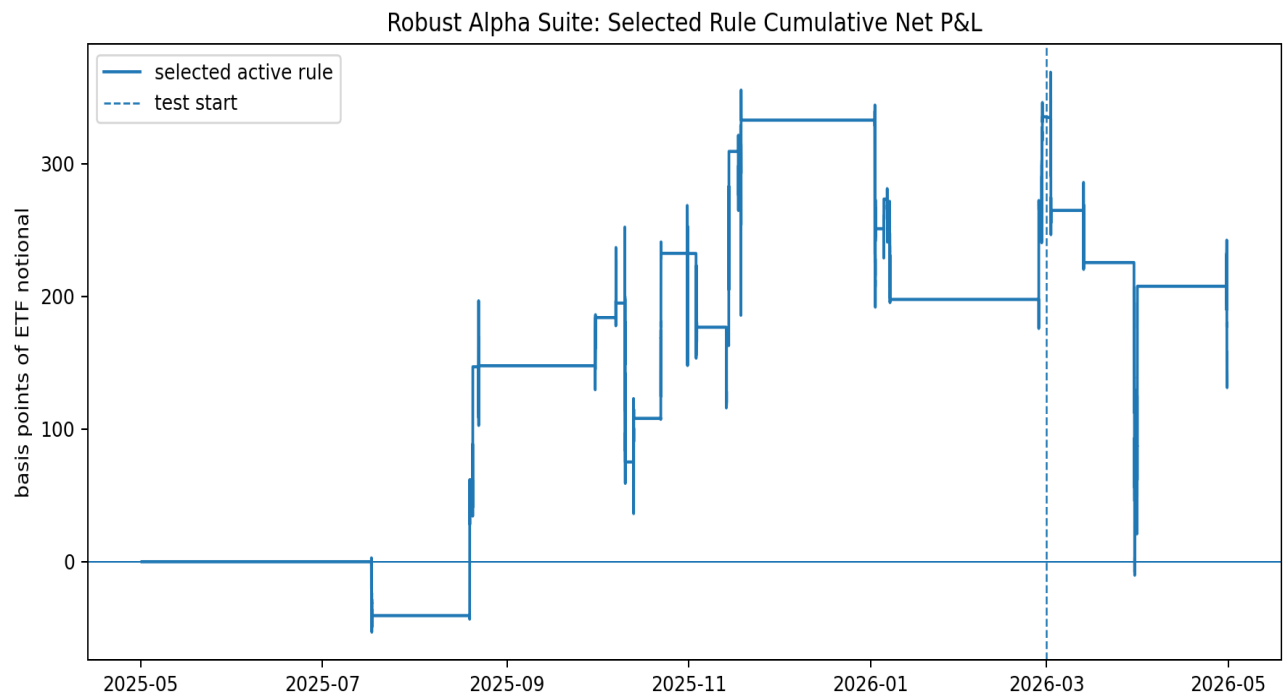
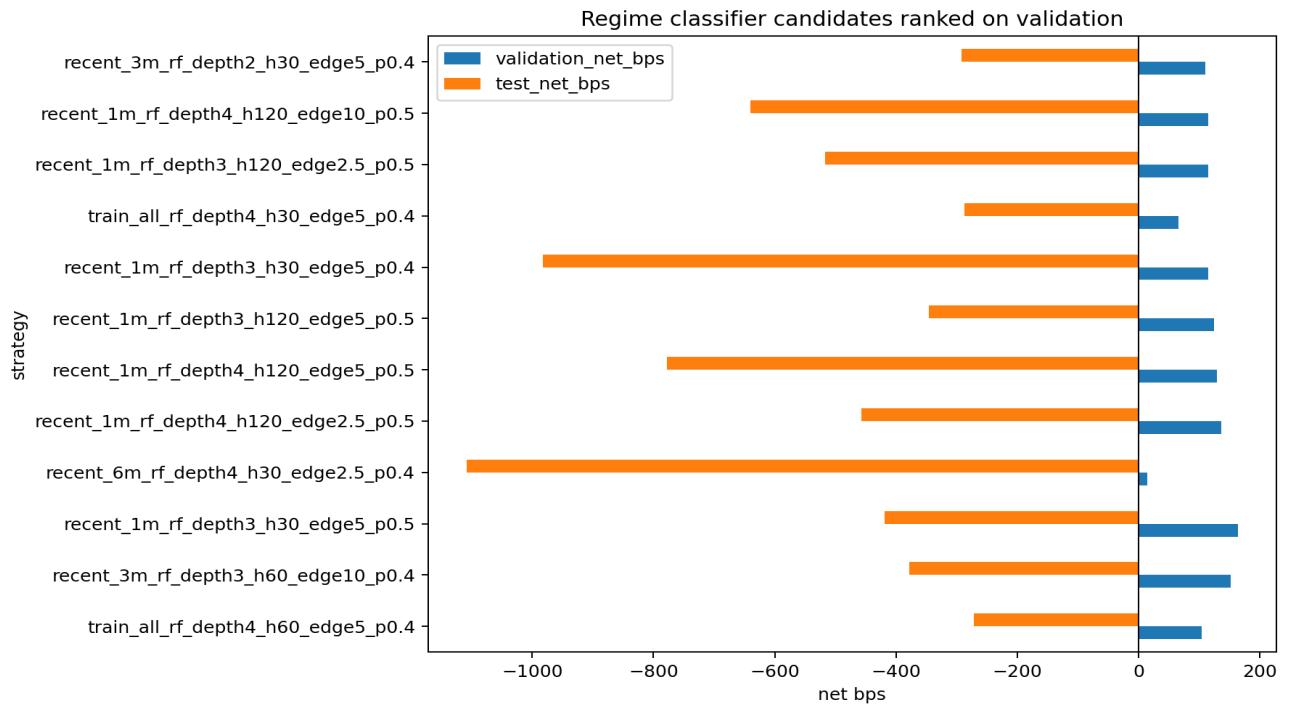
XLK-Only Timing

period	gross_bps	cost_bps	net_bps	trades
train	760.40	1795.16	-1034.76	386
validation	407.97	202.23	205.74	46
test	-550.25	302.18	-852.43	99
all	618.12	2299.57	-1681.45	531









Sparse-Basket Premium XLK Timing: Cumulative Net P&L

